

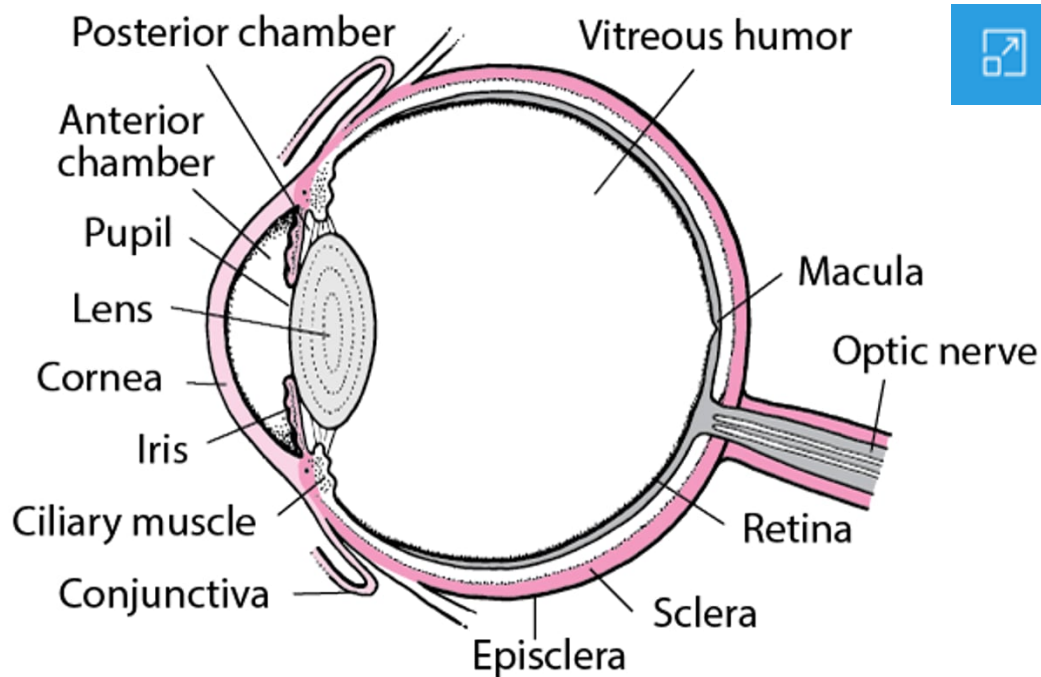
Sudden Vision Loss

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Loss of vision is considered sudden if it develops within a few minutes to a couple of days. It may affect one or both eyes and all or part of a field of vision. Loss of only a small part of the field of vision (for example, as a result of a small [retinal detachment](#)) may seem like [blurred vision](#). Other symptoms, for example [eye pain](#), may occur depending on the cause of vision loss.

An Inside Look at the Eye



Causes of Sudden Vision Loss

Sudden loss of vision has 3 general causes:

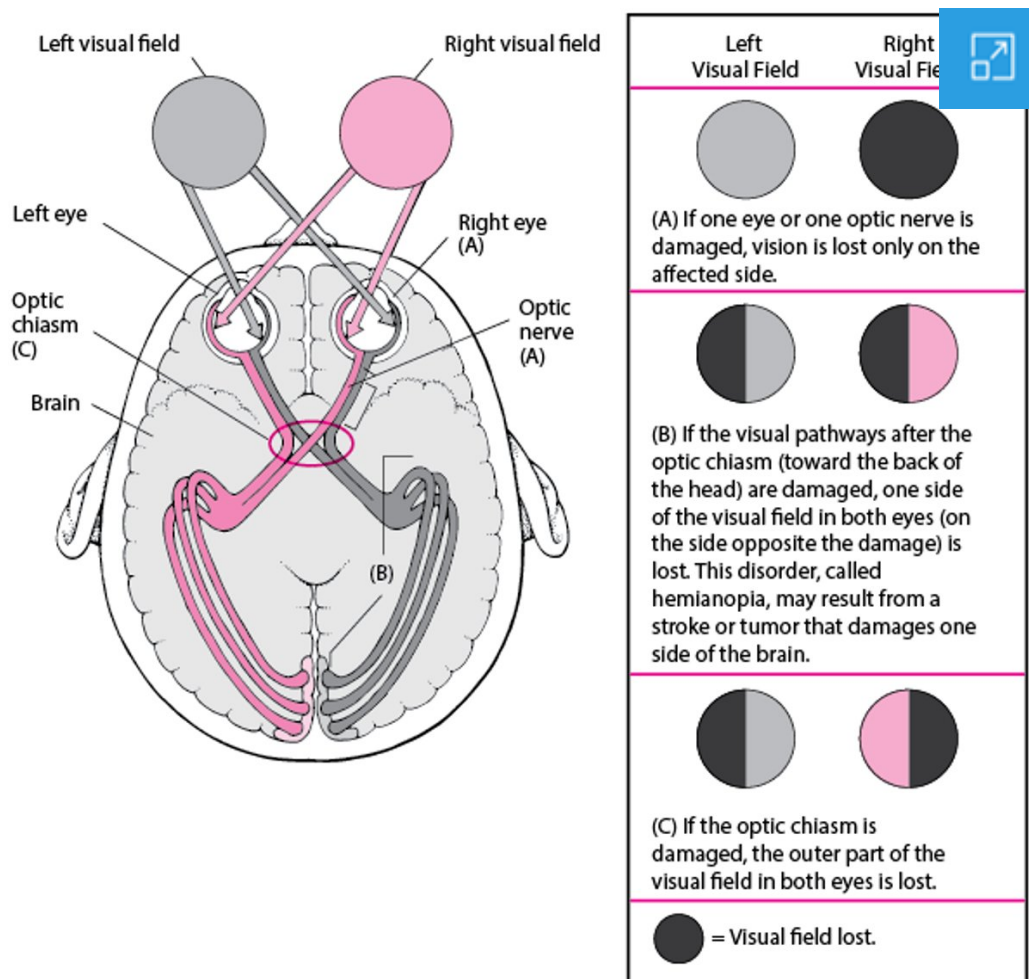
- Clouding of normally transparent eye structures
- Abnormalities of the retina (the light-sensing structure at the back of the eye)
- Abnormalities of the nerves that carry visual signals from the eye to the brain (the optic nerve and the visual pathways)

Light must travel through several transparent structures before it can be sensed by the retina. First, light passes through the cornea (the clear layer in front of the iris and pupil), then the lens, and then the vitreous humor (the jellylike substance that fills the eyeball). Anything that blocks light from passing through these structures, for example, a corneal ulcer or bleeding into the vitreous humor, or disrupts the transmission of nerve impulses from the back of the eye to the brain can cause loss of vision.

Most of the disorders that cause total loss of vision when they affect the entire eye may cause only partial vision loss when they affect only part of the eye.

When the Visual Pathways Are Damaged

Nerve signals travel along the optic nerve from each eye. The 2 optic nerves meet at the optic chiasm. There, the optic nerve from each eye divides, and half of the nerve fibers from each side cross to the other side. Because of this arrangement, the brain receives information via both optic nerves for the left visual field and for the right visual field. Damage to an eye or the visual pathway causes different types of vision loss depending on where the damage occurs.



Common causes

The most common causes of sudden, painless loss of vision are:

- Blockage of a major artery of the retina ([central retinal artery occlusion](#))
- Blockage of an artery to the optic nerve ([ischemic optic neuropathy](#))
- Blockage of a major vein in the retina ([central retinal vein occlusion](#))

- Blood in the jellylike vitreous humor near the back of the eye (vitreous hemorrhage)

Sudden retinal artery blockage can result from a blood clot or small piece of atherosclerotic material that breaks off and travels into the artery. The artery to the optic nerve can be blocked in the same ways and can also be blocked by inflammation (such as may occur with [giant cell \[temporal\] arteritis](#)). A blood clot can form in the retinal vein and block it, particularly in older adults with high blood pressure or diabetes. People with diabetes are also at risk of bleeding into the vitreous humor.

Sometimes what seems like a sudden start of symptoms may instead be sudden recognition. For example, a person with long-standing reduced vision in one eye (possibly caused by a dense cataract) may suddenly become aware of the reduced vision in the affected eye after covering the unaffected eye.

Less common causes

Less common causes of sudden loss of vision (see table

[Sudden Loss of Vision](#)) include [stroke](#) or [transient ischemic attack](#) (TIA), [acute glaucoma](#), [retinal detachment](#), inflammation of the structures in the front of the eye between the cornea and the lens (anterior [uveitis](#), sometimes called iritis), certain infections of the retina, and bleeding within the retina as a complication of [age-related macular degeneration](#).

How and Why Blindness Develops



Anything that blocks the passage of light from the environment to the back of the eye or disrupts the transmission of nerve impulses from the back of the eye to the brain will interfere with vision. In the United States, legal blindness is defined as a visual acuity of 20/200 or worse in the better eye, even after correction with eyeglasses or contact lenses, or a visual field restricted to less than 20° in the better eye. Many people who are considered legally blind can distinguish shapes and shadows but not normal detail.

Blindness can occur under the following circumstances:

Light cannot reach the retina.

- Damage to the cornea caused by infections such as herpes keratoconjunctivitis or an infection that follows contact lens overwearing, which results in an opaque corneal scar
- Damage to the cornea caused by vitamin A deficiency (keratomalacia), which causes dry eyes and results in an opaque corneal scar (rare in developed nations)
- Damage to the cornea caused by a severe injury that results in an opaque corneal scar
- A cataract, which causes loss of clarity of the lens

Light rays do not focus on the retina clearly.

- Imperfect focusing of light rays on the retina (refraction errors) that cannot be fully corrected with eyeglasses or contact lenses (such as from certain types of cataracts)

The retina cannot sense light rays normally.

- Detached retina
- Diabetes mellitus
- Macular degeneration
- Retinitis pigmentosa
- Inadequate blood supply to the retina, usually due to a blockage of the retinal artery or vein, which may be caused by inflammation of the blood vessel wall (such as that caused by giant cell arteritis), or due to a blood clot

that travels to the eye from somewhere else (such as from the carotid artery in the neck)

- Infection of the retina (such as from *Toxoplasma* or fungi)

Nerve impulses from the retina are not transmitted to the brain normally.

- Disorders affecting the optic nerve or its pathways inside the brain, such as brain tumors, strokes, infections, and multiple sclerosis
- [Glaucoma](#)
- Inflammation of the optic nerve ([optic neuritis](#))

The brain cannot interpret information sent by the eye.

- Disorders that affect the areas of the brain that interpret visual impulses (visual cortex), such as strokes and tumors



Some Causes and Features of Sudden Loss of Vision

Cause	Common Features*	Diagnostic Approach
Sudden loss of vision without eye pain		
Sudden, brief loss of vision in one eye resulting from a transient ischemic attack (called amaurosis fugax)	Blindness in one eye lasting minutes to hours	A doctor's examination MRI or CT angiography of the carotid arteries and brain Ultrasound of the carotid arteries Echocardiogram (ultrasound of the heart) ECG Continuous monitoring of heart rhythm
	Sometimes a normal eye examination	
Blockage of the central retinal artery (the artery that carries blood to the retina)	Almost instantaneous, complete loss of vision in one eye	A doctor's examination Measurement of ESR (a blood test), C-reactive protein, and platelets
	In people with risk factors for atherosclerosis (such as high blood pressure, abnormal blood lipids, or cigarette smoking)	Sometimes, MRI or CT of the brain Ultrasound of the carotid arteries Echocardiogram (ultrasound of the heart) ECG



MSD

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